Task 6: Automated Logging using Triggers & Views

A trigger that logs every INSERT or UPDATE

A view that shows daily activity reports

Real-Time Usage: Audit logging in enterprise databases.

Aim

To implement automated logging using triggers and generate daily activity reports using views in an order management system.

Algorithm

1. Use existing tables (customers, products, orders)
2. Create a log table to store activity details
3. Create triggers for:
    - INSERT on orders
    - UPDATE on orders
4. Automatically store logs when data changes
5. Create a view to display daily activity
6. Retrieve logs using SELECT from the view

Program:

```
USE order_db;

CREATE TABLE order_logs (

   log_id INT AUTO_INCREMENT PRIMARY KEY,

   order_id INT,

   action_type VARCHAR(20),

   action_time TIMESTAMP DEFAULT CURRENT_TIMESTAMP

);

DELIMITER $$

CREATE TRIGGER after_order_insert

AFTER INSERT ON orders
```

```
FOR EACH ROW
BEGIN
    INSERT INTO order_logs (order_id, action_type)
    VALUES (NEW.id, 'INSERT');
END$$
DELIMITER ;
DELIMITER $$

CREATE TRIGGER after_order_update
AFTER UPDATE ON orders
FOR EACH ROW
BEGIN
    INSERT INTO order_logs (order_id, action_type)
    VALUES (NEW.id, 'UPDATE');
END$$

DELIMITER ;
CREATE VIEW daily_activity AS
SELECT
    DATE(action_time) AS activity_date,
    action_type,
    COUNT(*) AS total_actions
FROM order_logs
GROUP BY DATE(action_time), action_type;
INSERT INTO orders (customer_id, product_id, quantity, order_date)
VALUES (1,1,2,'2024-05-01');
```

UPDATE orders SET quantity = 3 WHERE id = 1;

SELECT * FROM order_logs;

SELECT * FROM daily_activity;

Output:

Show query box

Current selection does not contain a unique column. Grid edit, checkbox, Edit, Copy and Delete features are not available.

Showing rows 0 - 1 (2 total, Query took 0.0056 seconds.)

```
SELECT * FROM daily_activity;
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all | Number of rows: 25 Filter rows: Search this table

Extra options

activity_date	action_type	total_actions
2026-04-15	INSERT	1
2026-04-15	UPDATE	1

Show all | Number of rows: 25 Filter rows: Search this table

Query results operations

Print Copy to clipboard Export Display chart Create view

Show query box

Current selection does not contain a unique column. Grid edit, checkbox, Edit, Copy and Delete features are not available.

Showing rows 0 - 1 (2 total, Query took 0.0171 seconds.)

```
SELECT * FROM daily_activity;
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all | Number of rows: 25 Filter rows: Search this table

Extra options

activity_date	customer_name	action_type	total_actions
2026-04-15	Preethi	INSERT	1
2026-04-15	Preethi	UPDATE	1

Show all | Number of rows: 25 Filter rows: Search this table

Query results operations

Print Copy to clipboard Export Display chart Create view

Result:

The system successfully logs database changes using triggers and generates daily activity reports using views. This ensures automated auditing and monitoring of data operations.